WSQ

FINANCIAL ANALYSIS

COURSE SLIDES · WSQ

Financial Analysis for Small and Medium Enterprises

WSQ Course Code: TGS-2026064860

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v13 · 11 August 2026

COURSE ADMINISTRATION

Welcome & Housekeeping

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer or administrator will show you the digital attendance QR code generated from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

About the Trainer

?

Your Trainer

General Trainer template — to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

YOUR TRAINER

About the Trainer

AA

Dr Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer, Tertiary Infotech Academy Pte Ltd

BACKGROUND
PhD; ACTA/ACLP-certified trainer with deep experience in data analytics, finance and business applications.

DELIVERS
WSQ courses on financial analysis, accounting systems, data analytics and business software.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

ICE-BREAKER

Let's Know Each Other

- Your name and organisation / role.
- Your experience with accounting or finance (if any).
- What you want to be able to analyse or decide after this course.

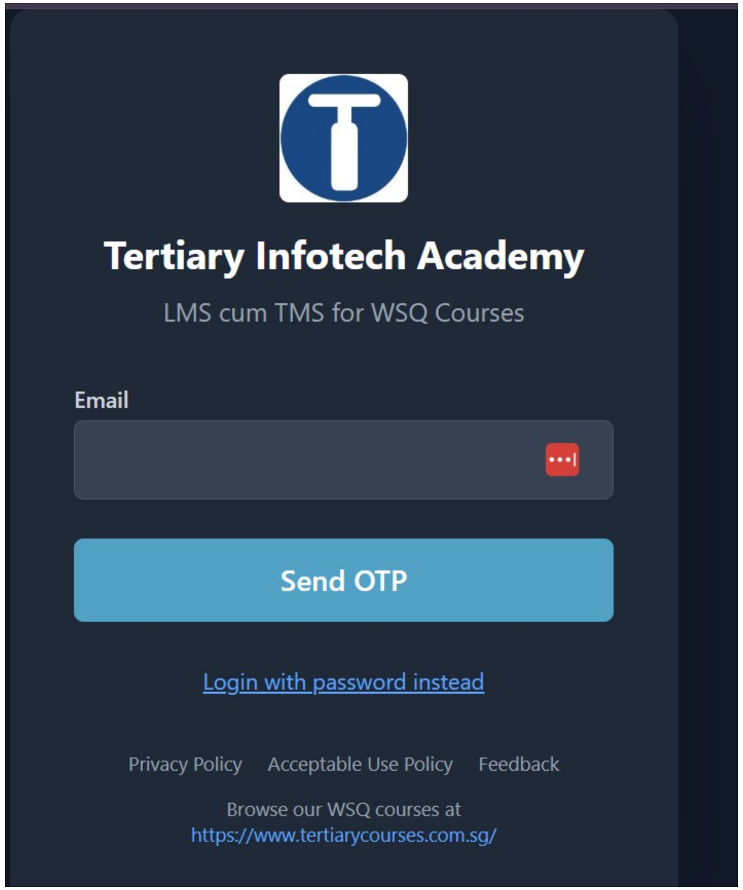
HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is stupid.
3. Mutual respect: agree to disagree.
4. One conversation at one time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

Download Course Material

- Go to https://lms-tms.tertiaryinfotech.com
- Log in with your registered email — an OTP is sent to you.
- Open your course: Financial Analysis for Small and Medium Enterprises
- Download the slides, Learner Guide and activity worksheets.
- Keep them handy — the final assessment is open book.



SCHEDULE

Lesson Plan — 2 Days, 8 Hours/Day

Day 1

- Day 1 — Financial Statements & Financial Ratios

– Digital Attendance (AM)

– Trainer and learner introductions · Learning outcomes · Course outline

– Topic 1: Understanding Financial Statements (Activity 1)

– Lunch break · Digital Attendance (PM)

– Topic 2: Analysing Financial Ratios (Activities 2–3)

Day 2

- Day 2 — Planning, Budgeting & Final Assessment

– Digital Attendance (AM)

– Topic 3: Planning & Budgeting using Financial Statements (Activities 4–6)

– Lunch break · Digital Attendance (PM)

– Revision · Course feedback & TRAQOM survey

– Digital Attendance (Assessment) · Final Assessment (WA + PP)

WSQ SKILLS FRAMEWORK

Skills Framework (TSC)

1

TSC Title · Code

Financial Analysis · ACC-MAC-5004-1.1

2

Knowledge K1–K2

Statement of financial position · Balance sheet

3

Knowledge K3–K4

Income and cash flow statements · Statement of changes in equity

4

Knowledge K5

Financial statement analysis techniques

5

Ability A1

Identify trends by comparing ratios across time periods and statement types

6

Ability A2

Prepare and interpret performance and position using financial statements

WHAT YOU'LL ACHIEVE

Learning Outcomes

1. **LO1 — Understand**
   Financial statements: the balance sheet, income statement and cash flow statement.

2. **LO2 — Evaluate**
   The organisation's financial performance from the trend of financial ratios.

3. **LO3 — Analyze**
   Financial statements and prepare the organisation's position.

COURSE OUTLINE

Course Outline — 3 Topics

Topic 1 (K1–K3)

- Overview of Finance and Chart of Accounts
- Balance Sheet Statement
- Profit and Loss (P&L) Statement
- Cash Flow Statement

Topic 2 (K4, A1)

- Ratios for Corporate Profitability
- Ratios for Corporate Performance
- Equity Changes Statement

Topic 3 (K5, A2)

- Analyse Financial Statements
- Financial Planning
- Capital Budgeting

Criteria for Funding

- Minimum attendance rate of 75% based on the SSG Digital Attendance record.
- Complete the assessment and be assessed as 'Competent'.

Briefing for Assessment

- Place phones and other materials under the table or on the floor.
- No photos or recording of assessment scripts.
- No discussion during the assessment.
- Use a black/blue pen for hard-copy assessments.
- No liquid paper or correction tape.
- Assessment scripts are collected when the time is up.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — financial-analysis tasks on real statements, 1 hour, open book.
- Format: Open Book — slides, Learner Guide and approved materials only.
- A minimum of 75% attendance is required to be eligible for assessment and funding.
- An appeal process is available if required.

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then PP — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

01

TOPIC 01

Understanding Financial Statements

Overview of Finance and Chart of Accounts · Balance Sheet Statement · Profit and Loss (P&L) Statement · Cash Flow Statement

TSC K1 · K2 · K3

Key Concepts — Understanding Financial Statements

1. **Finance & accounting**
   Why finance matters to every business decision — the language of the boardroom.

2. **Chart of accounts**
   The general-ledger account structure that every transaction is recorded against.

3. **Balance sheet**
   Assets = Liabilities + Equity — the statement of financial position at a point in time.

4. **Income statement**
   Revenue − Expenses = Profit — financial performance over an accounting period.

5. **Cash flow statement**
   Operating, investing and financing cash movements — where the cash actually went.

6. **Double-entry system**
   Debits equal credits, so the accounting equation always balances.

BUSINESS FINANCE

Finance is attaining money, investing it, and gaining a return on investment.

Any business decision has financial impact — finance is the common language of the boardroom.

WHY FINANCE MATTERS

Significance of Finance

1. Any business decision has financial impact.
2. It is the common language used in board meetings.
3. The more you understand finance, the more the insights.
4. Dealing with finance people becomes easy.
5. Relate finance to your own business unit.
6. Drive and achieve better performance.

The Finance Business Unit

1. **FP&A**
   Budgeting, forecasting, actual vs variance, management reports.

2. **Operations**
   Purchases and payables · expenses · sales and receivables.

3. **Controls**
   Internal controls, audit, processes and policies.

4. **Systems**
   Finance tools, transformation, ERP implementation, data & BI.

5. **M&A**
   Due diligence, checklists, M&A budget, pre/post-integration.

6. **Tax & Treasury**
   Corporate tax, GST, tax audit and cash flow management.

LENDERS ↔ BORROWERS

The Financial System

Lenders & regulators

- Regulators & institutions

– Ministry of Finance · ACRA · IRAS · SAC · CPF

– Banks · insurance companies · mutual funds · private equity funds

– Lenders provide capital to the system

Borrowers

- Borrowers & businesses

– Sole proprietorship · partnership / LLP

– Private company · public listed company

– Non-profit organisations · individuals · foreign companies

Accounting — the Big Picture

1 **Three questions**

How much money do I need? How do I raise it? How do I manage it?

2 **The equation**

Assets = Liabilities + Owner's Equity — always in balance.

3 **The mechanics**

Double entry · transaction analysis · summary of transactions.

4 **The outputs**

Income statement · owner's equity statement · balance sheet · statement of cash flows.

Objectives & Advantages of Accounting

Objectives

- Objectives

– Maintain a systematic record of transactions

– Ascertain profit or loss and financial position

– Assist management and communicate information to users

– Prevent manipulation and fraud

Advantages

- Advantages

– Financial information about the business; replaces memory

– Facilitates benchmarking, tax settlement and loans

– Evidence in court; facilitates sale of the business

– Helps in decision making

ACCOUNTING TYPES

Types of Accounting

1. **Financial accounting**
   Aggregates financial information into external reports — IFRS, GAAP.

2. **Management accounting**
   Internal operational reporting — cost accounting, payroll, payables.

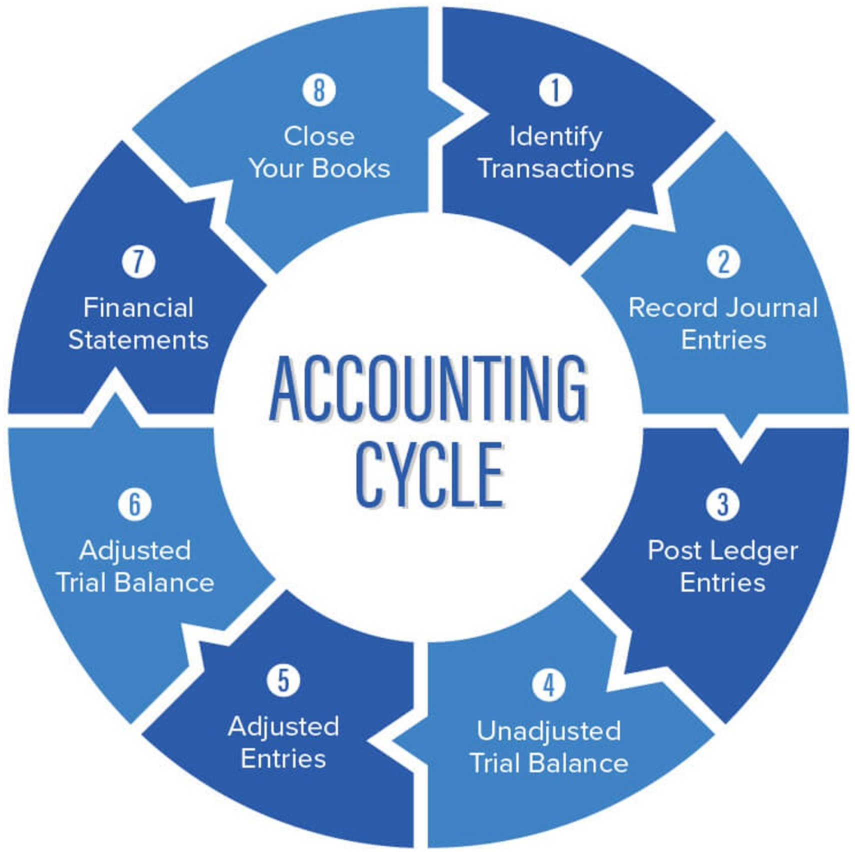
3. **Tax accounting**
   Compliance with tax regulations, filings and future tax planning.

4. **Auditing**
   Assurance that financial statements fairly present results and position.

5. **Forensic accounting**
   Reconstruction of financial information when records are incomplete.

6. **Internal auditing**
   Examination of systems and transactions to spot weaknesses, fraud and waste.

The Accounting Cycle



Identify transactions → journal entries → ledger → trial balance → adjustments → financial statements → close.

Accounting Periods

- An accounting period is the time span for which financial statements are prepared — e.g. a calendar year, a quarter or a month.
- The income statement and cash flow statement report amounts occurring DURING the period.
- The balance sheet reports assets and liabilities AS AT the final moment of the period.

Chart of Accounts

- A chart of accounts lists the account names a company uses to record transactions in its general ledger.
- Typical numbering: assets, liabilities, equity, revenue and expenses.
- Each account maps to either the balance sheet or the income statement.
- Companies tailor the chart to their needs and add accounts as required.

Chart of Accounts			
Number	Description	Account Type	Financial Statement
1-001	Cash	Asset	Balance Sheet
1-010	Accounts Receivable	Asset	Balance Sheet
1-020	Prepaid Expenses	Asset	Balance Sheet
1-030	Inventory	Asset	Balance Sheet
1-040	Fixed Assets	Asset	Balance Sheet
1-050	Accumulated Depreciation	Asset	Balance Sheet
1-060	Other Assets	Asset	Balance Sheet
2-001	Accounts Payable	Liability	Balance Sheet
2-010	Accrued Liabilities	Liability	Balance Sheet
2-020	Taxes Payable	Liability	Balance Sheet
2-030	Payroll Payable	Liability	Balance Sheet
2-040	Notes Payable	Liability	Balance Sheet
3-001	Common Stock	Equity	Balance Sheet
3-010	Retained Earnings	Equity	Balance Sheet
3-020	Additional Paid in Capital	Equity	Balance Sheet
4-001	Revenue	Revenue	Income Statement
4-010	Sales returns and allowances	Revenue	Income Statement
5-001	Cost of Goods Sold	Expense	Income Statement
5-010	Advertising Expense	Expense	Income Statement
5-020	Bank Fees	Expense	Income Statement
5-030	Depreciation Expense	Expense	Income Statement
5-040	Payroll Tax Expense	Expense	Income Statement
5-050	Rent Expense	Expense	Income Statement
5-060	Supplies Expense	Expense	Income Statement
5-070	Utilities Expense	Expense	Income Statement
5-080	Wages Expense	Expense	Income Statement
6-001	Other Expenses	Other	Income Statement

Single-Entry vs Double-Entry Accounting

Single entry

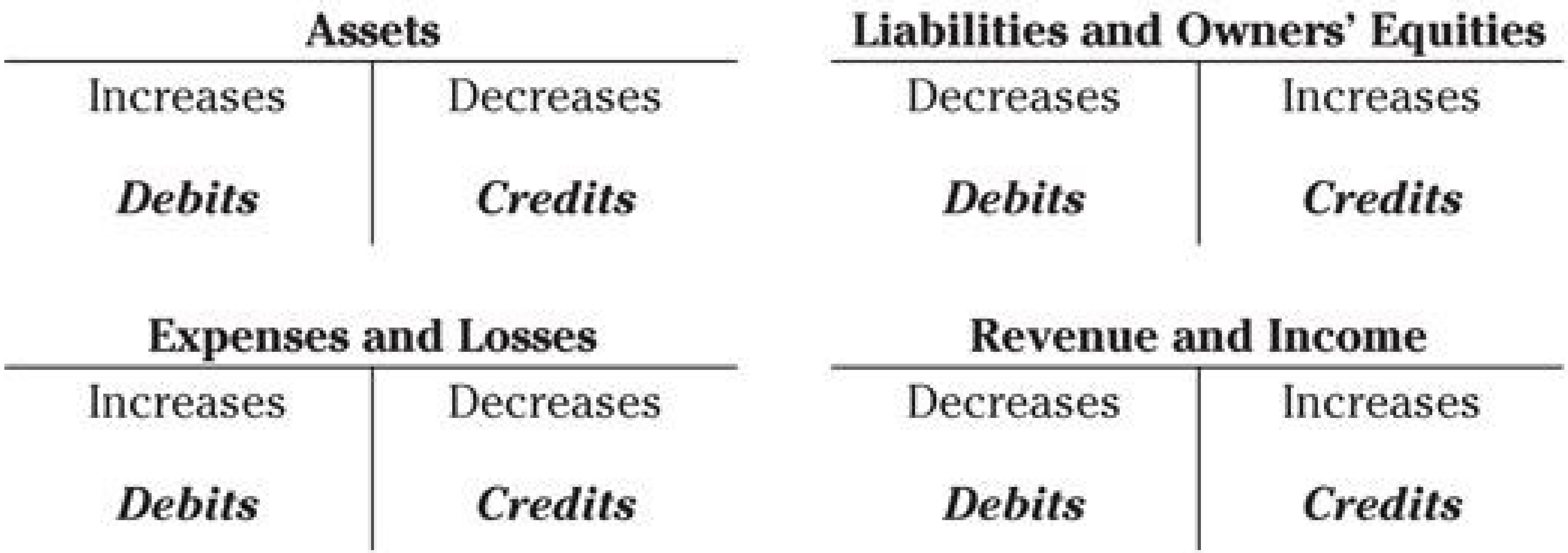
- Single entry
  - One entry per transaction, centred on the income statement
  - Tracks cash receipts and disbursements — the cash book
  - Assets and liabilities are not tracked systematically
  - Suitable only for the simplest businesses

Double entry

- Double entry
  - Every transaction hits at least two accounts
  - Debits always equal credits
  - The accounting equation stays in balance
  - The standard for every modern accounting system

WHICH SIDE INCREASES?

Double-Entry Rules — Debits & Credits



Assets and expenses increase with debits; liabilities, equity and revenue increase with credits.

Double-Entry Bookkeeping — Journal Example

Double Entry Bookkeeping

General Journal **#1001**

Date	Details	A/C #	Debit (Dr)	Credit (Cr)
1 May 2018	Cash	301	10,000	
	Sales	401		10,000
Sale of Inventory				

A $10,000 cash sale: debit Cash (A/C 301), credit Sales (A/C 401).

THE RECORDING DISCIPLINE

Bookkeeping

- Bookkeeping is the systematic gathering and recording of a company's financial transactions.
- Transactions are identified, approved, sorted and stored so they can be retrieved and presented in reports.
- Examples: purchase of supplies with cash · purchase/sale of merchandise on credit · rent · salaries and wages · buying equipment · borrowing from a bank.

The Three Key Financial Statements

Balance Sheet

- Statement of financial position
- Assets = Liabilities + Equity
- Liquidity and net worth
- A snapshot AS AT a date

Income Statement

- Profit & Loss statement
- Profits = Revenues − Expenses
- Profitability over a period
- Top line to bottom line

Cash Flow Statement

- Movement of cash in the business
- Operating · investing · financing
- Where cash came from and went
- Reconciles to the cash balance

WHO READS THE NUMBERS

Users of Financial Statements

1. **Management**
   Run the business — plans, budgets and performance reviews.

2. **Investors & owners**
   Assess returns, growth and the safety of their capital.

3. **Lenders & creditors**
   Judge the ability to repay loans and trade credit.

4. **Regulators & tax authorities**
   Compliance — ACRA filings, IRAS tax computations.

5. **Suppliers & customers**
   Decide whether to extend credit or rely on the firm.

6. **Employees**
   Job security, bonuses and profit-sharing.

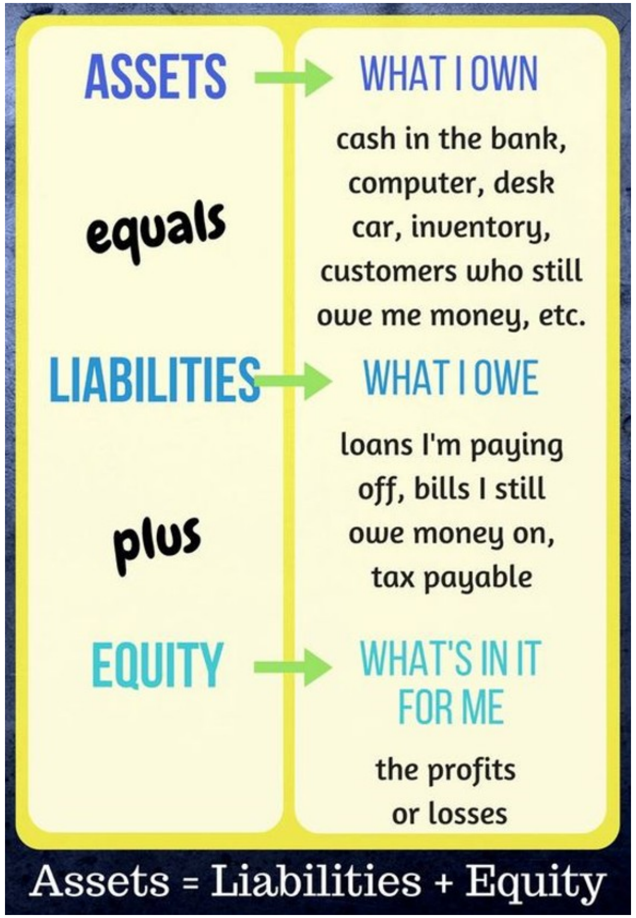
BALANCE SHEET · THE FUNDAMENTAL EQUATION

Assets = Liabilities + Owners' Equity

The balance sheet shows what the business owns (utilisation of funds) against what it owes and what the owners put in (sources of funds).

ASSETS = LIABILITIES + EQUITY

The Accounting Equation



Assets are what I own; liabilities are what I owe; equity is what is left for the owners.

THE BUILDING BLOCKS

Balance Sheet — Definitions

1

Assets

Resources that will provide a benefit in the future — utilisation of funds.

2

Liabilities

Obligations to repay money or provide a service in the future.

3

Owners' Equity

Money provided to the company by the owners.

4

Retained Earnings

Net income that is re-invested in the business.

STRUCTURE

Balance Sheet Components

Assets

- Assets

– Current assets: cash & equivalents, receivables, inventories, marketable securities

– Long-term assets: plant, property & equipment (PP&E), intangible assets

Liabilities & Equity

- Liabilities & Equity

– Current liabilities: accounts payable, current debt, current portion of LT debt

– Long-term liabilities: bonds payable, long-term debt

– Shareholders' equity: share capital + retained earnings

Balance Sheet — Example

XYZ COMPANY
Balance Sheet
12/31/2017

ASSETS	
Current Assets:	
Cash	$12,000
Accounts Receivable	35,000
Inventory	120,000
Prepaid Rent	8,000
Total Current Assets	$175,000
Long-Term Assets	
Land	$126,000
Buildings & Improvements	300,000
Furniture & Fixtures	50,000
General Equipment	125,000
Total Fixed Assets	$601,000
TOTAL ASSETS	**$776,000**
LIABILITIES	
Current Liabilities:	
Accounts Payable	$60,000
Taxes Payable	25,000
Salaries/Wages Payable	30,000
Interest Payable	25,000
Total Current Liabilities	$140,000
Long Term Liabilities:	
Loan 1	$322,000
Total Long Term Liabilities	$322,000
TOTAL LIABILITIES	**$462,000**
OWNER'S EQUITY	
Paid in Capital	$64,000
Retained Earnings	250,000
TOTAL OWNER'S EQUITY	**$314,000**
TOTAL LIABILITIES & OWNER'S EQUITY	**$776,000**

Current and long-term assets vs current and long-term liabilities and owners' equity.

Balance Sheet — Limitations

- The balance sheet is a static document — it depicts assets and liabilities 'as on date X' only.
- Assets are expressed at book value (historic cost less depreciation), which may differ from market value.
- Not all assets appear — e.g. internally-generated intangibles such as patents and a skilled workforce.

LIQUID · QUICK TO CONVERT TO CASH

Current Assets

1

Cash & equivalents

Legal tender, bills, coins, cheques; banker's acceptances, commercial paper, securities maturing < 90 days.

2

Accounts receivable

Funds owed by customers — recorded once goods or services are delivered.

3

Inventories

Raw materials, work-in-progress and finished goods held for sale.

4

Marketable securities

Common stock, treasury bills and bonds maturing within one year.

NON-CURRENT ASSETS

Long-Term Assets

1

PP&E

Plant, property and equipment — the tangible productive base of the business.

2

Depreciation

Spreads the asset's cost over time to match the income it produces; the method chosen affects results.

3

Intangible assets

Patents, trademarks, goodwill — non-physical assets with economic value.

4

Working capital

Current assets − current liabilities: the funds available to run daily operations.

Four Depreciation Methods

Straight line	**(Cost − Salvage) / Useful life**	$50,000 over 10 years → $5,000/yr
Double declining	**2 × SL rate × Book value**	Higher expense in early years
Units of production	**(Units / Life units) × (Cost − Salvage)**	3M of 100M units on $50,000 → $1,500
Sum of years	**(Remaining life / SYD) × (Cost − Salvage)**	$25,000, 8 yrs → yr 1 $5,556, yr 2 $4,861

OBLIGATIONS

Liabilities

Current (< 1 year)

- Current liabilities

– Accounts payable — money owed to vendors and suppliers

– Current debt / notes payable — borrowings payable within one year

– Current portion of long-term debt accrued this operating cycle

Long-term (> 1 year)

- Long-term liabilities

– Bonds payable — principal at a future date, periodic interest

– Bond types: serial, sinking fund, convertible, registered, secured, debenture

– Long-term debt — owed to creditors beyond 12 months

THE OWNERS' STAKE

Shareholders' Equity

1

Share capital

Money invested by the owners — common or preferred shares.

2

Capital stages

Authorized → issued → subscribed → called-up → paid-up capital.

3

Retained earnings

Net profits kept in the company after all dividends and distributions.

4

Private equity

Ownership interest in a private company.

INCOME STATEMENT · THE FUNDAMENTAL EQUATION

Profits = Revenues − Expenses

The income statement (Profit & Loss) shows financial performance across an accounting period, from the top line to the bottom line.

Income Statement — Structure

- **Revenue — the total of all sales (top line)**
  - − Cost of Goods Sold: direct materials, labour, overheads
- **= Gross Profit**
  - − Operating expenses: rental, admin payroll, utilities
- **= Operating Profit (EBIT)**
  - − Interest expenses on debt
- **= Profit before Tax → − Tax = Profit after Tax**

Example Corporation
Income Statement
For the year ended December 31, 2017

Sales (all on credit)	$500,000
Cost of goods sold	380,000
Gross profit	120,000
Operating expenses	
Selling expenses	35,000
Administrative expenses	45,000
Total operating expenses	80,000
Operating income	40,000
Interest expense	12,000
Income before taxes	28,000
Income tax expense	5,000
Net income after taxes	$ 23,000

COMPONENTS

Income Statement — Key Terms

1 **Revenue**
Money actually received in the period, net of discounts and returns — the top line.

2 **Expenses**
Economic costs incurred to earn revenue; tax-deductible expenses lower taxable income.

3 **EBITDA**
Earnings before interest, tax, depreciation and amortization — operating performance without financing/accounting effects.

4 **Depreciation & amortization**
Wear-and-tear of tangible assets (depreciation) and write-down of intangibles (amortization).

5 **Tax**
A mandatory financial charge imposed by the government on income.

6 **Net income**
What remains for the owners after every expense — the bottom line.

Summary of Income Statement Items

1. Revenue − COGS = Gross Profit
2. − Marketing, selling & admin = EBITDA
3. − Depreciation & amortization = EBIT
4. − Interest & other expenses = EBT
5. − Tax = Net Income

WHEN IS REVENUE RECOGNIZED?

Cash vs Accrual Basis Accounting

Accrual basis: recognise when the work is done. Cash basis: recognise when the cash is received.

FOLLOW THE CASH

Cash Flow Statement

1

What it shows

The movement of cash in the business — incoming and outgoing money.

2

Operating activities

Converts income-statement items from accrual to cash: sales income and production expenses.

3

Investing activities

Purchase and sale of long-term investments and PP&E — gains or losses on assets.

4

Financing activities

Issuance/repurchase of stock and bonds, loans and dividend payments.

WORKED EXAMPLE

Cash Flow Statement — Example

Finance Walk, Inc.
Statement of Cash Flows
For the Year ending Dec. 31, 20xx

Beginning Cash Balance (as of Jan. 1, 20xx)		$ 50,000.00
Operating Activities		
Sale of merchandise	$ 146,500.00	
Purchase of inventory	(35,000.00)	
Payment of accounts payable	(45,000.00)	
Net Cash Flow (used) provided by Operating Activities		66,500.00
Investing Activities		
Purchase of Equipment	$ (15,500.00)	
Sale of used furniture (at book value)	25,000.00	
Net Cash Flow (used) provided by Investing Activities		9,500.00
Financing Activities		
Additional bonds payable	$ 10,500.00	
Payment of long-term notes payable	(20,500.00)	
Dividends Paid	(19,500.00)	
Additional common stock investments	4,500.00	
Net Cash Flow (used) provided by Financing Activities		(25,000.00)
Ending Cash Balance (as of Dec. 31, 20xx)		**$ 101,000.00**

Operating, investing and financing blocks reconcile the opening cash balance to the closing balance.

Prepare a Cash Flow Statement

ACTIVITY 1

Havenwood Trading Pte Ltd is a fictitious Singapore SME distributor. From its FY2018/FY2019 balance sheets and FY2019 income statement (all in S$ million), derive the full statement of cash flows — operating, investing and financing — and reconcile it to the movement in cash.

You'll produce

A complete statement of cash flows (operating / investing / financing) for Havenwood Trading, derived from the balance sheet movements and the income statement.

Tools: Microsoft Excel · data workbook FA-Activity-01-Cash-Flow-Statement.xlsx in activities/activity01

All files: activities/activity01/ · **Detailed step-by-step: see this activity in the Learner Guide.**

TOPIC RECAP

Recap — Understanding Financial Statements

- The chart of accounts and double-entry rules keep the books in balance.
- The balance sheet shows position: Assets = Liabilities + Equity.
- The income statement shows performance: Profits = Revenues − Expenses.
- The cash flow statement shows where the cash actually moved.
- You prepared a full cash flow statement from a balance sheet and income statement.

02

TOPIC 02

Analysing Financial Ratios

Ratios for Corporate Profitability · Ratios for Corporate Performance · Equity Changes Statement

TSC K4 · A1

Key Concepts — Analysing Financial Ratios

1

Liquidity ratios

Current, quick, cash and operating-cash-flow ratios — can the firm pay its short-term bills?

2

Leverage ratios

Debt, debt-to-equity, interest coverage — how much risk is carried in the capital structure?

3

Efficiency ratios

Inventory, receivables, payables and asset turnover — how hard are the assets working?

4

Profitability ratios

Gross, operating, net margins, ROA, ROCE, ROE, EPS — is the business earning enough?

5

Equity changes statement

Reconciles opening to closing equity — share capital, dividends, retained earnings.

6

Trend analysis

Compare ratios across periods and statement types to spot direction and risk (A1).

THE ANALYST'S TOOLKIT

What is Financial Ratio Analysis?

- Financial ratios are mathematical indicators calculated by comparing key figures in the financial statements.
- They explain the reasons behind the current financial position and recent performance — and build expectations for the future.
- Example: net profit margin compares net income with net revenue — the dollars of profit earned per $100 of sales.
- Ratios only become meaningful against a benchmark: peers, the industry, or the company's own trend.

Identifying Relevant Data

STEP THREE: Select the data from the comparables' financials

Comparative Analysi	Apple	Google	Microsoft	Evaluation of the Compan
Profitability				
Net Margin (%)	22.8%	21.8%	13.0%	Higher Net Margins
Return on Assets (%)	20.4%	11.8%	7.0%	Higher ROA
Return on Equity (%)	46.2%	14.6%	14.4%	Higher ROE
Liquidity				
Current Ratio	1.1	4.7	2.5	Lower Liquidity
Quick Ratio	0.7	4.4	2.3	Lower Liquidity
Leverage				
Debt Ratio (%)	58.9%	18.4%	54.6%	Higher Leverage
Debt to Equity	44.8%	1.7%	34.7%	Higher Leverage
Efficiency				
Asset Turnover	0.89	0.54	0.54	Higher Asset Use
Receivables in Days	26.8	51.0	73.0	Better Collection Ability
Payables in Days	85.6	23.6	77.5	Better Short-Term Credit Ability
Market Value				
Price/Book Value	4.5	4.0	5.3	Lower Price/Book Value

Profitability, liquidity, leverage, efficiency and market-value ratios compared across companies.

WHY RATIOS

Benefits of Financial Ratios

1. Understand the profitability of the business.
2. Analyse operational efficiency.
3. Identify the liquidity of the business.
4. Identify business risks.
5. Identify financial risks.
6. Plan for the future.

RATIO MAP

The Four Ratio Families

Liquidity & Leverage

- Current, quick, cash ratios
- Operating cash flow ratio
- Debt & debt-to-equity ratios
- Interest & debt service coverage

Efficiency

- Asset turnover
- Inventory turnover
- Receivables & payables turnover
- Collection & payment periods

Profitability

- Gross / operating / net margins
- Return on assets & equity
- Return on capital employed
- Earnings per share

CAN THE FIRM PAY ITS BILLS?

Liquidity Ratios

Current Ratio	**Current Assets / Current Liabilities**	≥ 1 means current assets cover current bills
Quick (Acid Test) Ratio	**(Cash + Securities + Receivables) / CL**	Most readily available assets only
Cash Ratio	**(Cash + Cash Equivalents) / CL**	Normal value is below 1.00
Operating Cash Flow Ratio	**Operating Cash Flow / CL**	Cash generated vs short-term obligations

INTERPRETATION

Reading the Liquidity Ratios

- Current ratio ≥ 1: current assets exceed current liabilities — no expected liquidity problem; below 1 signals possible stress.
- Quick ratio strips out inventory; a rule of thumb is 0.5, but always compare with the industry.
- A quick ratio below the industry average may mean difficulty honouring current obligations.
- Creditors like a high cash ratio — but idle cash earns nothing, so firms typically hold less than 1.00.

Quick Ratio — Worked Example

Quick Ratio= (Current Assets-Inventory) / Current Liabilities

Item	Company A	Company B
Quick Assets	$100,000	$170,000
Current Liabilities	$105,000	$140,000
Current Ratio	**0.95**	**1.21**

Quick assets over current liabilities: Company A 0.95 vs Company B 1.21.

Leverage Ratios

Debt Ratio	**Total Debt / Total Assets**	< 0.5 stable · > 0.5 stability issue
Debt-to-Equity	**Total Liabilities / Shareholders' Equity**	1.00 = half debt-financed; lower is safer
Times Interest Earned	**EBIT / Interest Expense**	Can earnings cover the interest bill?
Fixed Charge Coverage	**(EBIT + Lease pmts) / (Interest + Lease pmts)**	Adds lease obligations to the test
Debt Service Coverage	**Operating Income / Total Debt Service**	Principal + interest coverage

INTERPRETATION

Reading the Leverage Ratios

- Debt ratio measures the risk that assets cannot pay off debts — critical for long-term sustainability.
- Very low debt can also mean under-utilised financing and restricted growth.
- A rising debt-to-equity trend is a warning: more of the assets are financed by lenders.
- Higher times-interest-earned is favourable — the company earns enough to pay its interest.

HOW HARD ARE THE ASSETS WORKING?

Efficiency Ratios (1)

Asset Turnover	**Net Sales / Average Total Assets**	Revenue per dollar of assets
Inventory Turnover	**COGS / Average Inventories**	High = efficient; low = over-stocking
Receivables Turnover	**Net Credit Sales / Average AR**	How fast credit sales are collected
Payables Turnover	**Net Purchases / Average AP**	How quickly suppliers are repaid

TURNOVER & DAYS

Efficiency Ratios (2)

Fixed Asset Turnover	**Net Revenue / Average Fixed Assets**	Revenue per dollar of fixed assets
Working Capital Turnover	**Revenue / Average Working Capital**	Revenue per dollar of working capital
Average Collection Period	**365 / Receivables Turnover**	AR $40k on sales $400k → 36.5 days
Average Payment Period	**365 / Payables Turnover**	AP turnover 2 → 182.5 days to pay

COMPANY A VS COMPANY B

Inventory Turnover — Worked Example

Inventory Turnover Ratio = Cost of Goods Sold / Average Inventory

Item	Company A	Company B
Cost of Goods Sold	$70,000	$105,000
Average Inventory	$30,000	$60,000
Inventory Turnover Ratio	**2.33**	**1.75**

COGS over average inventory: Company A turns 2.33× vs Company B 1.75×.

IS THE BUSINESS EARNING ENOUGH?

Profitability Ratios

Gross Profit Margin	**Gross Profit / Revenue**	Production efficiency / pricing strategy
Operating Margin	**Operating Income (EBIT) / Revenue**	9% → $0.09 profit per $1 of sales
Net Profit Margin	**Net Income / Net Sales**	The most basic profitability measure
Return on Assets	**Net Income / Average Total Assets**	Cents earned per dollar of assets
Return on Capital Employed	**Net Operating Profit / Capital Employed**	Includes long-term finance
Return on Equity	**Net Income / Average Shareholders' Equity**	The owners' return on new investment

INTERPRETATION

Reading the Profitability Ratios

- Higher ratios are more favourable — they reflect the ability to generate earnings.
- Compare only within the same industry: asset-heavy industries naturally show lower ROA.
- A rising ROA / ROE trend means profitability is improving; a falling trend means it is deteriorating.
- ROE can be manipulated with debt — never rely on ROE alone for investment decisions.

Net Profit Ratio — Worked Example

Net Profit Ratio= Earnings Before Interest & Tax / Sales

Item	Company A	Company B
Net Profit	$36,000	$52,000
Net Sales	$140,000	$210,000
Net Profit Ratio	**26%**	**25%**

Net profit over net sales: Company A 26% vs Company B 25%.

Earnings Per Share (EPS)

- **EPS = (Net Income − Preferred Dividends) / Weighted Average Common Shares**
- Standardises earnings by the number of shares outstanding.
- Not comparable alone — different companies have different share counts.
- Feeds the price-to-earnings (P/E) ratio used by investors.

	A	B	C	D	E	F
19						
20		Net Income	450000			
21		Preferred Dividends	30000			
22		Weighted Average Number of Common Shares	70000			
23						
24		Earnings per Share	6	per share	=(C20-C21)/C22	
25						

PUTTING IT TOGETHER

Measuring Profitability — Sample Company

- **Gross Margin: 37.50%**
- **Operating Margin: 26.67%**
- **Net Profit Margin: 16.67%**
- **Return on Equity: 9.09%**
- **Return on Assets: 4%**
- Derived from the sample balance sheet and income statement shown.

ple Balance Sheet of ıpany A)	
	200
	300
	500
	30
i	250
	280
y	100
	120
	220

Summary of C (All figures in '0C
Sales
Cost of Goods Sc
Gross Profit
Marketing Exper
General Admin E
Operating Profit
Interest Expense
Income Before T
Taxes
Net Income

Statement of Changes in Equity

1

What it is

Also called the statement of retained earnings — reconciles opening to closing equity.

2

The equation

Beginning Equity + Net Income − Dividends ± Other changes = Ending Equity.

3

Components

Opening balance · policy changes · prior-period corrections · restated balance.

4

More components

Changes in share capital · dividends · income/loss · revaluation reserve · other gains and losses.

ONE-PAGE CHEAT SHEET

Summary of All Ratios

Liquidity & Leverage

- Liquidity
  - Current = CA / CL
  - Acid test = (CA − Inventories) / CL
  - Cash = Cash & equivalents / CL
  - Operating CF = OCF / CL
- Leverage
  - Debt = Total liabilities / Total assets
  - Debt-to-equity = Total liabilities / Equity
  - Debt service coverage = Op. income / Debt service

Efficiency & Profitability

- Efficiency
  - Asset turnover = Net sales / Avg total assets
  - Inventory turnover = COGS / Avg inventory
  - Receivables turnover = Net sales / Avg AR
  - Payables turnover = Net purchases / Avg AP
- Profitability
  - Gross margin = Gross profit / Net sales
  - Operating margin = EBIT / Net sales · Net margin = NI / Net sales
  - ROE = NI / Equity · ROA = NI / Total assets

Ratio Analysis of Two Companies

ACTIVITY 2

Orchid Logistics Pte Ltd (Company X) and Marina Retail Group Pte Ltd (Company Y) are two fictitious SMEs. From their FY2025 balance sheets and income statements, compute the liquidity, profitability, turnover and solvency ratios for both companies and give a one-line verdict per ratio family on which company performs better.

You'll produce

A completed ratio-analysis worksheet comparing Company X and Company Y across four ratio families, with a one-line verdict per ratio.

Tools: Microsoft Excel · data workbook FA-Activity-02-Ratio-Analysis.xlsx in activities/activity02

All files: activities/activity02/ · **Detailed step-by-step: see this activity in the Learner Guide.**

Trend Analysis over Four Years

ACTIVITY 3

Sunrise F&B Holdings Pte Ltd is a fictitious cafe-chain SME. From four years (FY2018–FY2021) of balance sheets and income statements, compute year-on-year growth for every line, derive the key ratios for each year, and present your observations on where the business is heading.

You'll produce

A completed trend-analysis template: YoY growth for every balance-sheet and income-statement line plus a four-year ratio table with your observations.

Tools: Microsoft Excel · data workbook FA-Activity-03-Trend-Analysis.xlsx in activities/activity03

All files: activities/activity03/ · **Detailed step-by-step: see this activity in the Learner Guide.**

TOPIC RECAP

Recap — Analysing Financial Ratios

- Liquidity, leverage, efficiency and profitability — the four ratio families.
- Every ratio needs a benchmark: industry, competitor or the company's own trend.
- The statement of changes in equity reconciles opening to closing equity.
- You compared two companies ratio-by-ratio and ran a four-year trend analysis.

03

TOPIC 03

Planning & Budgeting using Financial Statements

Analyse Financial Statements · Financial Planning · Capital Budgeting

Key Concepts — Planning & Budgeting using Financial Statements

1 **Budgeting**
Baseline, incremental, zero-based and hybrid budgets — planning income vs spending.

2 **Capital budgeting**
Payback, discounted payback, NPV, PI and IRR — evaluating long-term investments.

3 **Time value of money**
$PV = FV / (1+i)^n$ — a dollar today is worth more than a dollar tomorrow.

4 **Forecast & variance**
Budget vs actual vs forecast — favourable and adverse variances, thresholds.

5 **Financial health**
Read the balance sheet, income and cash flow statements together, then ratio-check.

6 **Analysis methods**
Ratio, horizontal (trend) and vertical analysis, plus industry benchmarking (A2).

BUDGET

Budgeting is balancing your expenses with your income —

by creating a plan to spend your money. To balance a budget you have two levers: increase income or decrease spending.

EXPENSES · INCOME

Key Components of a Budget

1 **Fixed expenses**
Stay the same month to month — e.g. rent payments.

2 **Flexible expenses**
Change month to month — e.g. utilities.

3 **Capital vs operating**
Capital expenses buy long-term assets; operating expenses run the business.

4 **Total income**
Income from operations and other sources — investments, rentals and more.

5 **Disposable income**
What is left after taxes are subtracted.

6 **Variance**
The gap between the plan and the actual outcome.

FOUR APPROACHES

Broad Budgeting Methods

1

Baseline budget

Begins with a previous plan as the baseline.

2

Incremental budget

A % or $ increment on the previous baseline.

3

Zero-based budget

Starts fresh, like a brand-new plan.

4

Hybrid budget

A combination of any of the above.

TERMS YOU WILL MEET

Budgeting Vocabulary

1. Fiscal year · budget communications
2. Budget targets · budget guidelines
3. Annual plan · budget review
4. Board review · approved plan
5. Forecast · actual
6. Variance

Budgeting — Objectives

- Provides clear structure, guidance and a basis for the action plan.
- Predicts expected cash flows and allocates resources to the right activities.
- Lets management prioritise projects and model scenarios — worst, median, best and custom cases.
- Measures actual performance against the approved plan and analyses the causes of variances.
- Ensures control over spending and mandates responsibility and accountability.

Budgeting — Advantages & Success Factors

Advantages

- Advantages
  - Planning orientation and profitability review
  - Assumptions review and performance evaluation
  - Funding planning and cash allocation
  - Bottleneck analysis

Success factors

- Success factors
  - Management support and employee involvement
  - Clear organisational goals and responsibility accounting
  - Sound accounting system and organisational structure
  - Flexibility and communication of results

The Budgeting Process

Stakeholder input → goal setting → needs assessment → direction → operating budgets → adoption → monitoring → adjustment.

Sample Budget Report

QBO Advanced Training

BUDGET OVERVIEW: BUDGET FY 2019 - DRAFT V2 - FY19 P&L

January - December 2019

	JAN 2019	FEB 2019	MAR 2019	APR 2019	MAY 2019	JUN 2019	JUL 2019	AUG 2019	SEP 2019	OCT 2019	NOV 2019	DEC 2019	TOTAL
▾ Revenue													
Sales of Product Revenue	173,133.56	59,154.14	60,000.00	60,000.00	60,000.00	60,000.00	60,000.00	60,000.00	60,000.00	60,000.00	60,000.00	60,000.00	S$832,287.70
Total Revenue	**S$173,133.56**	**S$59,154.14**	**S$60,000.00**	**S$60,000.00**	**S$60,000.00**	**S$60,000.00**	**S$60,000.00**	**S$60,000.00**	**S$60,000.00**	**S$60,000.00**	**S$60,000.00**	**S$60,000.00**	**S$832,287.70**
▾ Cost of Sales													
Cost of sales	90,000.00	26,000.00	30,000.00	30,000.00	30,000.00	30,000.00	30,000.00	30,000.00	30,000.00	30,000.00	30,000.00	30,000.00	S$416,000.00
Total Cost of Sales	**S$90,000.00**	**S$26,000.00**	**S$30,000.00**	**S$30,000.00**	**S$30,000.00**	**S$30,000.00**	**S$30,000.00**	**S$30,000.00**	**S$30,000.00**	**S$30,000.00**	**S$30,000.00**	**S$30,000.00**	**S$416,000.00**
GROSS PROFIT	**S$83,133.56**	**S$33,154.14**	**S$30,000.00**	**S$30,000.00**	**S$30,000.00**	**S$30,000.00**	**S$30,000.00**	**S$30,000.00**	**S$30,000.00**	**S$30,000.00**	**S$30,000.00**	**S$30,000.00**	**S$416,287.70**
▾ Expenses													
Admin Expenses	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	S$24,000.00
Advertising and Promotion	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	1,000.00	S$12,000.00
Bank Service Charges	300.00	300.00	300.00	300.00	300.00	300.00	300.00	300.00	300.00	300.00	300.00	300.00	S$3,600.00
Computer and Internet Expen...	150.00	150.00	150.00	150.00	150.00	150.00	150.00	150.00	150.00	150.00	150.00	150.00	S$1,800.00
Depreciation Expense	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	S$24,000.00
Insurance Expense	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00	S$6,000.00
Meals and Entertainment	200.00	200.00	200.00	200.00	200.00	200.00	200.00	200.00	200.00	200.00	200.00	200.00	S$2,400.00
Office Supplies	300.00	300.00	300.00	300.00	300.00	300.00	300.00	300.00	300.00	300.00	300.00	300.00	S$3,600.00
Payroll Expenses	12,000.00	12,000.00	12,000.00	12,000.00	12,000.00	12,000.00	12,000.00	12,000.00	12,000.00	12,000.00	12,000.00	12,000.00	S$144,000.00
Postage & Courier	140.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	S$140.00
Rent Expense	3,000.00	3,000.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	S$6,000.00
▾ Salaries & Wages	2,500.00	2,500.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	S$5,000.00
Employer CPF	425.00	425.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	S$850.00
Total Salaries & Wages	**2,925.00**	**2,925.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**S$5,850.00**
Telephone Expense	300.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	S$300.00
Travel Expense	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	S$24,000.00
Total Expenses	**S$26,815.00**	**S$26,375.00**	**S$20,450.00**	**S$20,450.00**	**S$20,450.00**	**S$20,450.00**	**S$20,450.00**	**S$20,450.00**	**S$20,450.00**	**S$20,450.00**	**S$20,450.00**	**S$20,450.00**	**S$257,690.00**
NET OPERATING INCOME	**S$56,318.56**	**S$6,779.14**	**S$9,550.00**	**S$9,550.00**	**S$9,550.00**	**S$9,550.00**	**S$9,550.00**	**S$9,550.00**	**S$9,550.00**	**S$9,550.00**	**S$9,550.00**	**S$9,550.00**	**S$158,597.70**
▾ Other Expenses													
Exchange Gain or Loss	-1,993.89	-570.38	41.68	-7.59	0.00	0.00	0.00	0.00	1,210.26	0.00	0.00	0.00	S$ -1,319.92
Total Other Expenses	**S$ -1,993.89**	**S$ -570.38**	**S$41.68**	**S$ -7.59**	**S$0.00**	**S$0.00**	**S$0.00**	**S$0.00**	**S$1,210.26**	**S$0.00**	**S$0.00**	**S$0.00**	**S$ -1,319.92**
NET OTHER REVENUE	**S$1,993.89**	**S$570.38**	**S$ -41.68**	**S$7.59**	**S$0.00**	**S$0.00**	**S$0.00**	**S$0.00**	**S$ -1,210.26**	**S$0.00**	**S$0.00**	**S$0.00**	**S$1,319.92**
NET INCOME	**S$58,312.45**	**S$7,349.52**	**S$9,508.32**	**S$9,557.59**	**S$9,550.00**	**S$9,550.00**	**S$9,550.00**	**S$9,550.00**	**S$8,339.74**	**S$9,550.00**	**S$9,550.00**	**S$9,550.00**	**S$159,917.62**

A monthly phased budget: revenue and expense lines with totals for the fiscal year.

Capital Budgeting

1

Purpose

Planning for long-term investments in projects and assets.

2

Cash, not profit

Focus on cash inflows and outflows, not accounting revenue and expenses.

3

Time value of money

Adjust future cash flows to today's dollars before comparing.

4

Decision methods

Payback · discounted payback · NPV · profitability index · IRR.

A DOLLAR TODAY BEATS A DOLLAR TOMORROW

Time Value of Money

Present Value	**PV = FV / (1 + i)^n**	\$10,000 in 2 yrs @5% → \$9,070.30 today
Future Value	**FV = PV × (1 + i)^n**	\$10,000 today @5% → \$11,025 in 2 yrs
Components	**PV · FV · N periods · Interest i · Payment PMT**	Opportunity cost: interest or inflation

SIX STEPS

Capital Budgeting Process

1. Identify long-term goals
2. Identify investment proposals
3. Estimate & analyse the cash flows
4. Analyse financial feasibility
5. Choose & implement projects
6. Monitor the projects

FIVE DECISION TOOLS

Capital Budgeting Methods

Payback Period	Outlay / Annual cash inflow	$2,000 / $375 = 5.33 years
Discounted Payback	Cumulative discounted CF ≥ outlay	Same project at 10% → ≈ 8 years
Net Present Value	Σ PV(inflows) − Outlay	Accept if NPV > 0
Profitability Index	PV(inflows) / PV(outflows)	Accept if PI > 1
Internal Rate of Return	Rate where NPV = 0	Accept if IRR > threshold rate

DISCOUNTING THE CASH FLOWS

Discounted Payback — Worked Example

Year	Cash Flow	Discounted Cash Flow	Cumulative	Cash Flow	Discounted Cash Flow	Cumulative	Cash Flow	Discounted Cash Flow	Cumulative
0	-$1,000			-$1,000			-$1,000		
1	$250	$227	$227	$350	$318	$318	$500	$455	$455
2	$250	$207	$434	$350	$289	$607	$500	$413	$868
3	$250	$188	$622	$350	$263	$870	$500	$376	***$1,243***
4	$250	$171	$792	$350	$239	***$1,109***			
5	$250	$155	$948	$350	$217	$1,327			
6	$250	$141	***$1,089***						
7	$250	$128	$1,217						
8	$250	$117	$1,334						
9	$250	$106	$1,440						
10	$250	$96	$1,536						

Each future inflow is discounted before accumulating — payback arrives later than the simple method.

PI > 1 → ACCEPT

Profitability Index — Worked Example

- Discount every inflow at the required return.
- PV of inflows $2,304 vs PV of outflows $2,000.
- **PI = 2,304 / 2,000 = 1.15 → accept the project.**
- NPV = $304 — the same accept signal.

Year	Cash Flow	Discounted Cash Flows (i=5%)	Discounted cash Flows (i=3%)
0	-$10,000	-$10,000	-$10,000
1	$1,000	$952	$971
2	$1,000	$907	$943
3	$1,000	$864	$915
4	$1,000	$823	$888
5	$1,000	$784	$863
6	$1,500	$1,119	$1,256
7	$1,500	$1,066	$1,220
8	$1,500	$1,015	$1,184
9	$1,500	$967	$1,150
10	$1,500	$921	$1,116
	PV of cash inflows	$9,418	$10,505
	PV of Cash Outflow	$10,000	$10,000
	PI	94%	105%

Payback Period & Discounted Payback Period

ACTIVITY 4

Havenwood Trading is evaluating a refrigerated delivery van: S$2,000k outlay in Year 0, saving S$375k per year for 10 years versus outsourced cold-chain delivery. Compute the simple payback period, then discount each cash flow at the 10% required return to find the discounted payback.

You'll produce

A payback worksheet showing cumulative cash flow per year, the simple payback point (5.33 years) and the discounted payback point (≈8 years).

Tools: Microsoft Excel · data workbook FA-Activity-04-Payback-Period.xlsx in activities/activity04

All files: activities/activity04/ · **Detailed step-by-step: see this activity in the Learner Guide.**

Net Present Value & Profitability Index

ACTIVITY 5

Using the same delivery-van proposal (S$2,000k outlay, S$375k/year for 10 years, 10% required return), compute the Net Present Value and the Profitability Index, and make the accept/reject recommendation to management.

You'll produce

An NPV/PI worksheet: PV of each inflow, NPV ≈ S$304k, PI ≈ 1.15, and a reasoned accept decision.

Tools: Microsoft Excel · data workbook FA-Activity-05-NPV-PI.xlsx in activities/activity05

All files: activities/activity05/ · **Detailed step-by-step: see this activity in the Learner Guide.**

MONTHLY CASH DISCIPLINE

Cash Flow Analysis & Management

- Track the monthly statement of cash flows.
- Improve cash flow: invoice promptly and chase receivables.
- Negotiate supplier terms; manage inventory tightly.
- Time capital spending; keep a cash buffer.

ABC Cleaning Service				
Statement of Cash Flows				
For the month ended January 31, 2020				
CASH AND CASH EQUIVALENTS, BEGINNING OF PERIOD				$ 50,000.00
OPERATING ACTIVITIES				
	Net Income			$ 6,475.00
	ADJUST FOR OPERATING ACTIVITIES:			
	Depreciation on fixed assets			$ 1,000.00
	Decrease in current assets:			
	- Accounts receivable			$ (150.00)
	- Inventory			$ (19,900.00)
	- Prepaid expenses			$ (500.00)
	Increase in current liabilities:			
	- Accounts payable			$ 24,500.00
	- Deferred revenue			$ 725.00
	NET CASH FROM OPERATING ACTIVITIES			$ 12,150.00

Pro Forma (Projected) Financial Statements

1

Purpose

Plan ahead and compare actual vs predicted performance.

2

What is projected

The income statement and the balance sheet.

3

Cash flow

The projected cash flow is derived from the other two statements.

4

Forecasting cash

Estimate monthly sales → predict receipt of payments → estimate costs.

Budget, Actual, Variance, Forecast

- Budget: planned sales of $500,000. Actual: sales of $400,000.
- Variance: −$100,000 — or −20% of budget ($100,000 / $500,000).
- Both the dollar and percentage variance appear on the Budget-to-Actual report.
- Forecast: the projection of future periods from historical data, updated at regular intervals.

PLAN VS PREDICTION

Budget vs Forecast

Budget

- Budget

– A financial plan expressed in quantitative terms, prepared in advance

– The financial expression of a business target

– Sets targets · updated annually

– Variance analysis: yes

Forecast

- Forecast

– Estimation of future trends from past and present data

– A prediction of upcoming events and trends

– No targets · updated at regular intervals

– Variance analysis: no

ADVERSE VS FAVOURABLE

Types of Variances

- Adverse variance: actual income below budget, or spending above budget — a deficit.
- Favourable variance: actual income above budget, or spending below budget — a surplus.
- Manage over/underspend with clear-cut thresholds in dollars or percent.
- RAG-status thresholds trigger review at defined limits.

Dimension	RAG Status	Guidelines
Schedule	Green	Schedule is on target.
	Amber	There are likely to be minor delays in the schedule up to 30 days.
	Red	There will be delays of greater than 30 days.
Scope	Green	Scope is in line with agreed business case.
	Amber	There are likely to be minor changes in scope to original business case.
	Red	There will be a significant change in the scope delivered by the project.
Cost	Green	Cost is on target
	Amber	There is likely to be a cost overrun to the originally agreed budget by up to 10% or <£50,000 or and underrun of up to 10% or <£100,000
	Red	There is highly likely there will be a cost overrun to the originally agreed budget by > 10% or <£50,000 and underrun of > 10% or <£100,000
Benefits	Green	Benefits are on target.
	Amber	There is likely to be a reduction in benefits of up to 10% or £50,000
	Red	It is highly likely that there will be a reduction in benefits of >10% or £100,000

RISK PROFILING

Business & Financial Risk

1 **High-risk company**
Operates in a high-risk industry AND faces a risk of financial failure.

2 **High-risk factors**
Financial risk (banks, merchant accounts) · compliance · safety and statutory law.

3 **Consequences**
High-risk borrowers pay higher loan rates and larger down payments.

4 **Risk types**
Strategic · compliance · financial · operational · reputational risks.

5 **Examples**
Agriculture, construction and mining industries.

6 **Low-risk company**
Stable industry and sound finances — the reverse conditions.

THREE-STATEMENT REVIEW

Determine the Financial Health of a Company

Balance sheet

- Debt relative to equity
- Short-term liquidity
- Tangible vs financial assets
- Collection & repayment cycles
- Time to sell inventory

Income statement

- Revenue growth by period
- Gross profit margin
- Net profit % of revenue
- Interest cover on debt
- Payout vs reinvestment

Cash flow statement

- Liquidity situation
- Sources of cash
- Free cash flow generated
- Overall cash increase or decrease

RATIO CHECKLIST

…then Ratio-Check the Health

1 **Margins**
Gross profit margin and net profit margin on revenue.

2 **Coverage**
Ability to meet obligations — debt and interest payments.

3 **Liquidity**
Current ratio and quick ratio for obligations under one year.

4 **Leverage**
Debt-to-equity: the financing mix.

5 **Efficiency**
Inventory turnover and total asset turnover.

6 **Returns**
ROE and ROA: profit from equity and from assets.

WHAT INVESTORS EXAMINE

Financial Analysis for Investment Suitability

1

Revenues

Quantity, quality and timing · growth (ignore one-offs) · concentration · revenue per employee.

2

Profits

Gross, operating and net profit margins.

3

Operational efficiency

Receivables turnover and inventory turnover.

4

Market data

P/E ratio · price-to-book · PEG ratio.

5

Capital & solvency

Return on equity · debt-to-equity · liquidity.

6

DuPont index

Net Profit Margin × Asset Turnover × Equity Multiplier = ROE decomposed.

SECTOR-SPECIFIC RATIOS

Industry Ratios — Banking & Insurance

Banking

- Banking

– Net interest margin = (Interest income − expense) / Total assets

– Efficiency ratio = Non-interest expense / Revenue

– Liquidity coverage ratio · leverage ratio · CET1 ratio

Insurance

- Insurance

– Persistency = paying policyholders / active policyholders

– Solvency = available / required solvency margin

– Combined ratio · incurred claims ratio · claim settlement ratio

AGAINST INDUSTRY AVERAGES

Benchmarking Financial Performance

Financial ratio	2021	2020	2019	2018	2017	2016
Solvency Ratios						
Debt ratio	0.57	0.61	0.61	0.62	0.63	0.65
Debt-to-equity ratio	0.93	1.07	1.12	0.87	0.80	0.75
Interest coverage ratio	3.04	1.22	2.20	1.98	1.84	1.69
Liquidity Ratios						
Current Ratio	2.03	1.94	1.69	1.58	1.56	1.53
Quick Ratio	1.25	1.25	1.08	1.05	1.09	1.09
Cash Ratio	0.82	0.82	0.51	0.40	0.42	0.41
Profitability Ratios						
Profit margin	5.2%	1.5%	4%	3.3%	2.6%	1.9%
ROE (Return on equity), after tax	2.6%	-2.1%	2.7%	0.7%	-1.1%	-2.5%
ROA (Return on assets)	1.1%	-0.1%	1%	0.8%	0.5%	0.2%
Gross margin	41.6%	41.5%	42.2%	42.2%	41.7%	41%
Operating margin (Return on sales)	9.7%	4.8%	8%	7.4%	6.4%	5.7%
Activity Ratios						
Asset turnover (days)	815	869	747	654	635	636
Receivables turnover (days)	48	54	50	48	48	48
Inventory turnover (days)	76	79	71	71	70	71
Price Ratios						
Dividend Payout Ratio	0.27	0.27	0.32	0.27	0.31	0.29

Trend of US sectoral averages — compare your ratios against the sector, not in isolation.

THREE LENSES

Financial Statement Analysis — Methods

Ratio analysis

- Generate the ratios
- Compare against benchmarks
- Track the ratio trend over time

Horizontal (trend)

- Compare across periods
- % change of the same line item
- Spot growth and deterioration

Vertical analysis

- Every line as % of a base figure
- IS lines as % of gross sales
- BS lines as % of total assets

Horizontal Analysis — Example

Village Shipping Inc. Income Statement Horizontal Analysis For the years ending December 31, 2014 and December 31, 2015	2014	2015	% Change 2015 from 2014
Sales	500,000	475,000	-5.00%
Cost of goods sold	269,000	265,000	-1.49%
Gross profit	231,000	210,000	-9.09%
Wages	163,000	154,000	-5.52%
Repairs	4,150	5,800	39.76%
Rent	12,000	13,000	8.33%
Taxes	17,930	16,940	-5.52%
Office expenses	587	1,023	14.14%
Total expenses	197,667	190,763	-3.49%
Net Income	33,333	19,237	-42.29%

Income statement lines for two years with the percentage change of each item.

FOUR-YEAR BALANCE SHEET

Horizontal Analysis — Case Study

ITEM	YEAR			
Assets	2021	2020	2019	2018
Current Assets				
Cash	$30,000	$30,000	$30,000	$30,000
Inventory	$40,000	$34,000	$32,000	$30,000
Accounts Receivable	$88,180	$81,648	$75,600	$70,000
Fixed Assets				
Property	$194,000	$196,000	$198,000	$200,000
Others	$39,998	$39,999	$40,000	$40,000
Total	**$392,178**	**$381,647**	**$375,600**	**$370,000**

Liabilities				
Current Liabilities				
Accounts Payable	$80,000	$60,000	$51,000	$50,000
Short Term Loans	$44,090	$40,824	$37,800	$35,000
Long Term Liabilities	$50,000	$50,000	$50,000	$50,000
Capital and Retained Earnings	$218,088	$230,823	$236,800	$235,000
Total	**$392,178**	**$381,647**	**$375,600**	**$370,000**

Balance-sheet items tracked across 2018–2021 — the raw material for trend analysis.

EVERY LINE AS % OF THE BASE

Vertical Analysis — Example

ABC Company Vertical Analysis

	Year 1	Year 2	Year 3
Sales	100%	100%	100%
Cost of Goods Sold	30%	29%	40%
Gross Profit	**70%**	**71%**	**60%**
Salaries	24%	23%	30%
Rent	6%	6%	5%
Marketing	5%	5%	10%
Utilities	1%	1%	1%
Other expenses	2%	2%	2%
Total Expenses	**38%**	**37%**	**48%**
Net Income	**32%**	**34%**	**12%**

Income-statement items expressed as a percentage of sales for three years.

Solvency & Financial Risk Analysis

ACTIVITY 6

Return to Orchid Logistics (Company X) and Marina Retail Group (Company Y). Using their balance sheets and income statements, compare the two companies' solvency risk — debt-to-equity, financial leverage and interest coverage — and conclude which is the riskier borrower.

You'll produce

A solvency comparison table (D/E, leverage, interest coverage) for X and Y with a reasoned risk verdict.

Tools: Microsoft Excel · data workbook FA-Activity-06-Solvency-Analysis.xlsx in activities/activity06

All files: activities/activity06/ · **Detailed step-by-step: see this activity in the Learner Guide.**

TOPIC RECAP

Recap — Planning & Budgeting using Financial Statements

- Budgets plan the spending; forecasts predict the outcome; variances close the loop.
- Capital budgeting decisions: payback, discounted payback, NPV, PI and IRR.
- Financial health = balance sheet + income statement + cash flow, then ratio-check.
- Ratio, horizontal and vertical analysis — plus benchmarking — complete the toolkit.
- You evaluated an investment project and compared two companies' solvency risk.

WRAP-UP

Course Summary & Next Steps

LEARNING OUTCOMES

What You Achieved

1 **Financial statements (LO1)**
Read and prepared balance sheets, income statements and cash flow statements.

2 **Ratio analysis (LO2)**
Evaluated performance with liquidity, leverage, efficiency and profitability ratios and their trends.

3 **Planning & budgeting (LO3)**
Built budgets, evaluated investments (payback, NPV, PI) and assessed financial health and risk.

KEEP LEARNING

Recommended Courses

- WSQ - Quickbooks Accounting System for Small and Medium Enterprises
- WSQ - Unlocking the Power of Accounting and Tax Systems for SMEs
- WSQ - Statistical Data Analysis with Excel for Beginners
- WSQ - Budgeting for Small and Medium Enterprises
- WSQ - Software Automation with Excel VBA Programming

WE'RE HERE TO HELP

Support

- If you have any enquiries during and after the class, contact us:
- Email: enquiry@tertiaryinfotech.com
- Tel: +65 6100 0613
- Website: www.tertiarycourses.com.sg

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — financial-analysis tasks on real statements, 1 hour, open book.
- Open book: slides, Learner Guide and approved materials only.
- Remember to take the Assessment digital attendance (TRAQOM).
- Submit your completed answers on the LMS at https://lms-tms.tertiaryinfotech.com/.

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then PP — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

https://lms-tms.tertiaryinfotech.com/

Cert & TRAQOM Survey (Mandatory)

Certificate Delivery

Scan to receive your certificate

https://goo.gl/R2eumq

TRAQOM Survey

Your feedback helps us improve

https://ssgtraqom.qualtrics.com/jfe/form/SV_3K9i7rTJ9OLsauW?Q_CHL=qr

Scan the QR codes to receive your certificate and complete the TRAQOM survey.

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer or administrator will show you the digital attendance QR code generated from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

SEE YOU IN THE NEXT COURSE

Thank You!

You can now read, analyse and plan with financial statements — and put them to work for your SME.